

Figure 7.48 Hypermetropic eye and correction

greater than 25 cm. The positive sign of the focal length suggests that the correcting lens should be a convex lens. In principle, the convex lens slightly converges the rays coming from an object beyond y and makes them fall at the retina.

7.6.6.3 Astigmatism

Astigmatism is the defect arising due to different curvatures along different planes in the eye lens. Astigmatic person cannot see in all the directions equally well. The defect due to astigmatism is more serious than myopia and hypermetropia. The astigmatism can be corrected using a lens which has different curvatures in different planes. In general, these specially made glasses with different curvature for different planes are called as cylindrical lenses.

Due to aging people may develop combination of more than one defect. If it is the combination of nearsightedness and farsightedness then, such persons may need a converging glass for reading purpose and a diverging glass for seeing at a distance. Bifocal lenses and progressive lenses provide solution for these kinds of problems.

EXAMPLE 7.22

Calculate the power of the lens of the spectacles needed to rectify the defect of nearsightedness for a person who could see clearly up to a distance of 1.8 m.

Solution

The maximum distance the person could see is, $x = 1.8$ m.

The lens should have a focal length of, $f = -x$ m $= -1.8$ m.

It is a concave (or) diverging lens.

The power of the lens is,

$$P = -\frac{1}{1.8\ \text{m}} = -0.56\ \text{D}.$$

EXAMPLE 7.23

A person has farsightedness with the far distance he could see clearly is 75 cm. Calculate the power of the lens of the spectacles needed to rectify the defect.

Solution

The minimum distance the person could see clearly is, $y = 75$ cm.

The lens should have a focal length of,

$$f = \frac{y \times 25\ \text{cm}}{y - 25\ \text{cm}}$$

$$f = \frac{75\ \text{cm} \times 25\ \text{cm}}{75\ \text{cm} - 25\ \text{cm}} = 37.5\ \text{cm}$$

It is a convex (or) converging lens.

The power of the lens is,

$$P = \frac{1}{0.375\ \text{m}} = 2.67\ \text{D}.$$

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